Assignment 3
40 + 10 points

Stochastic Calculus for Finance

Due Sunday, March 21

Note: If you are using Python, it is recommended to submit all your programs in a Jupyter notebook. If you are using a different programming language, you can have multiple programs for each of the questions below. Make sure the different files are clearly labelled. Please employ clean code writing practices so that your code is comprehensible to others. Think of this as a trial run for the thousands of programs you would have to write as a quant.

Each of the ten exercises below is for 4 points, for a total of 40 points. You will get an additional 10 points for writing reasonably clean code.

In this assignment, we explore the fair pricing of a derivative security based on the binomial asset pricing model by numerical simulation. We recall important facts about the binomial asset pricing model. The price of a stock is modelled as a random variable on the sample space of N coin tosses with empirical coin toss probabilities p, q. A sequence of N coin tosses is written as $\underline{\omega} = \omega_1 \cdots \omega_N$ with each ω_i equal to H or T depending on the outcome of the i^{th} coin toss. The stock price process $\mathbb{S}_n$ is determined by the following evolution equation:

$$\mathbb{S}_{n+1} = \begin{cases} u\mathbb{S}_n & \omega_{n+1} = H \ , \\ d\mathbb{S}_n & \omega_{n+1} = T \ . \end{cases} \tag{1}$$

This shows that the stock price at time $\mathcal{S}_n$ for every $0 \leq n \leq N$ is a random variable on the sample space of coin tosses. Moreover, the stock price at time n depends only on the first n coin tosses.

We now discuss the pricing of an option based on the stock $\mathbb{S}_n$. For a European call option with strike price K and expiration time N, the payoff at time N is given by $\mathbb{V}_N = v_N(\mathbb{S}_N)$ where v_N is the following ordinary function

$$v_N(x) = \max(x - K, 0) = (x - K, 0)^+ \ . \tag{2}$$

The fair price of the option $\mathbb{V}_n$ at a prior time n is given by the conditional expectation

$$\frac{\mathbb{V}_n}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N} \,\middle|\, \mathcal{A}_n\right] \ , \tag{3}$$

where r is the risk-free interest rate and $\widetilde{\mathbb{E}}$ is the expectation taken with respect to the risk-neutral probabilities $\tilde{p}$, $\tilde{q}$. The initial price of the option is then

$$\mathbb{V}_0 = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N}\right] \ . \tag{4}$$

Recall how the risk-neutral probability measure comes about. First we set up a hedge for a short position on the derivative security by looking at a portfolio made up of stock and money market investments. This is a random process $\mathbb{X}_n$, whose evolution is given by

$$\mathbb{X}_{n+1} = \mathbb{\Delta}_n\mathbb{S}_{n+1} + (1+r)(\mathbb{X}_n - \mathbb{\Delta}_n\mathbb{S}_n) \ . \tag{5}$$

We want to choose the number of shares $\mathbb{\Delta}_n$ of the stock at each time n so that the value of the portfolio at time N is equal to the payoff of the option: $\mathbb{X}_N(\underline{\omega}) = \mathbb{V}_N(\underline{\omega})$ for all $\underline{\omega} \in \Omega_N$, the sample space of N coin tosses. That is, irrespective of the outcome of the coin tosses and the resulting stock price, we want our hedge to neutralize the risk involved in trading in the option. In other words, we want our portfolio to reproduce an average rate of return equal to the risk-free interest rate r of the money market.

Is there a probability measure $\tilde{p}$, $\tilde{q}$ (for the coin toss outcomes H and T respectively) such that the average or mean rate of return for the hedging portfolio after one period is the risk-free interest rate r? If we take conditional expectation of the wealth equation (5) with respect to $\mathcal{A}_n$ under some probability measure $\tilde{p}$, $\tilde{q}$ (which we want to be risk-neutral eventually), we get

$$\widetilde{\mathbb{E}}[\mathbb{X}_{n+1} \mid \mathcal{A}_n] = \mathbb{\Delta}_n \widetilde{\mathbb{E}}[\mathbb{S}_{n+1} \mid \mathcal{A}_n] + (1+r)(\mathbb{X}_n - \mathbb{\Delta}_n \mathbb{S}_n) \ . \tag{6}$$

Since we demand the average rate of return after one period to be equal to $(1+r)\mathbb{X}_n$, setting the above conditional expectation to this value, we get

$$(1+r)\mathbb{X}_n = \mathbb{\Delta}_n \widetilde{\mathbb{E}}[\mathbb{S}_{n+1} \mid \mathcal{A}_n] + (1+r)(\mathbb{X}_n - \mathbb{\Delta}_n \mathbb{S}_n) \ , \tag{7}$$

which, after cancelling some terms, becomes

$$\widetilde{\mathbb{E}}[\mathbb{S}_{n+1} \mid \mathcal{A}_n] = (1+r)\mathbb{S}_n \ . \tag{8}$$

In other words, under the probability measure $\tilde{p}$, $\tilde{q}$, the stock price process must have a mean rate of return equal to the risk-free interest rate r of the money market. Then we are guaranteed to get a mean rate of return equal to the risk-free interest rate r for our hedging portfolio $\mathbb{X}_n$. Solving the above equation using the binomial model for the stock price (1), we get the risk-neutral probabilities

$$\tilde{p} = \frac{1+r-d}{u-d} \ , \quad \tilde{q} = \frac{u-(1+r)}{u-d} \ . \tag{9}$$

Note that the above probabilities are both positive only if $d < 1+r < u$. This is also the condition that ensures that there are no arbitrage opportunities in the stock and money market.

The equation (8) involving stock prices can be rephrased as "the discounted stock price $\frac{1}{(1+r)^n}\mathbb{S}_n$ is a martingale under the risk-neutral probabilities". The conclusion is that an effective hedging portfolio will have value $\mathbb{X}_n$ at time n determined by the martingale condition under the risk-neutral probability measure:

$$\frac{\mathbb{X}_n}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{\mathbb{X}_N}{(1+r)^N} \,\middle|\, \mathcal{A}_n \right] \ . \tag{10}$$

The fair option price at each time n is then set equal to the value of the hedging portfolio, $\mathbb{V}_n = \mathbb{X}_n$, which gives

$$\frac{\mathbb{V}_n}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N} \,\middle|\, \mathcal{A}_n\right] \ . \tag{11}$$

To proceed further, let us summarize what we learnt in Assignment 2 through some tedious calculations. We learnt, by looking at some small values for m, n with $m < n$, that the conditional expectation $\mathbb{E}[\mathbb{M}_n \mid \mathcal{A}_m]$ for $m < n$ is the same as the conditional expectation $\mathbb{E}[\mathbb{M}_n \mid \sigma(\mathbb{M}_m)]$ and is equal to $\mathbb{M}_m$ for a symmetric random walk:

$$\mathbb{E}[\mathbb{M}_n \mid \mathcal{A}_m] = \mathbb{E}[\mathbb{M}_n \mid \sigma(\mathbb{M}_m)] = \mathbb{M}_m \ . \tag{12}$$

Similarly, we showed that

$$\mathbb{E}[f(\mathbb{M}_n) \mid \mathcal{A}_m] = \mathbb{E}[f(\mathbb{M}_n) \mid \sigma(\mathbb{M}_m)] = g(\mathbb{M}_m) \ , \tag{13}$$

where $g(x)$ is an ordinary function determined in terms of $f(x)$. These results actually follow in one stroke from the fact that $\mathbb{M}_n$ is a Markov process (See the solutions for Assignment 2). We derived these results in Assignment 2 without the knowledge that $\mathbb{M}_n$ is a Markov process. However, the Markov nature of $\mathbb{M}_n$ manifested itself in the conditional probabilities that were used in calculating the above conditional expectations, which is why we got what we got.

The result for a general Markov process $\mathbb{X}_n$ that is adapted to a filtration $\{\mathcal{A}_n\}$ is

$$\mathbb{E}[f(\mathbb{X}_n) \mid \mathcal{A}_m] = \mathbb{E}[f(\mathbb{X}_n) \mid \sigma(\mathbb{X}_m)] = \mathbb{E}[f(\mathbb{X}_n) \mid \mathbb{X}_m] = g(\mathbb{X}_m) \ . \tag{14}$$

The stock price process $\mathbb{S}_n$, with up and down factors u and d respectively, is also Markov since it is simply an exponentiation of the random walk process $\mathbb{M}_n$. Based on the above discussion, we can now write

$$\mathbb{E}[f(\mathbb{S}_N) \mid \mathcal{A}_n] = g(\mathbb{S}_n) \ , \quad \text{for } 0 \le n \le N, \tag{15}$$

where f is any ordinary non-random function of its argument and g is an appropriate non-random function obtained by computing the conditional expectation on the left hand side. The important point is that the right hand side is a function of only the stock price $\mathbb{S}_n$ at time n and not of the full stock price path up to time n.

Now, suppose we have an option whose payoff $\mathbb{V}_N$ at expiration time N is only a function of the final stock price $\mathbb{S}_N$. That is, $\mathbb{V}_N = v_N(\mathbb{S}_N)$ for some function v_N. For instance, this is the case for a European call or put option. We then have

$$\frac{\mathbb{V}_n}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{v_N(\mathbb{S}_N)}{(1+r)^N} \,\middle|\, \mathcal{A}_n\right] = \frac{v_n(\mathbb{S}_n)}{(1+r)^n} \ , \tag{16}$$

where $v_n(x)$ is an ordinary function of x which is determined by computing the conditional expectation value above. This follows from the general formula (15). This implies the strong result that, if the payoff of the option is a function only of the stock price at expiry time N, the fair price of the option $\mathbb{V}_n$ at time n depends on the time n and only the price of the stock $\mathbb{S}_n$ at time n.

There are two important features in the above formula (16):

1. The probability measure is with respect to the risk-neutral probabilities. This came about by demanding that the hedging portfolio has the same average rate of return as the risk-free interest rate r of the money market.
2. The fact that the price of the option $\mathbb{V}_n$ at time n is dependent only on the stock price $\mathbb{S}_n$ at time n through the function $v_n(\mathbb{S}_n)$. This came about directly from the fact that the payoff of the option is a function of the final stock price and that the stock price process $\mathbb{S}_n$ is a Markov process.

If the payoff of the option at expiration time is not a function of only the final stock price but is dependent on the full stock price path (e.g. exotic options like the Asian option or a lookback option), then one does not have (16). One has to fall back to the original formula (3).

We showed in Assignment 2 that a process of the type $(1+r)^{-n} v_n(\mathbb{S}_n)$ defined as above is a martingale. That is,

$$\frac{v_n(\mathbb{S}_n)}{(1+r)^n} = \widetilde{\mathbb{E}}\left[\frac{v_{n+1}(\mathbb{S}_{n+1})}{(1+r)^{n+1}} \,\middle|\, \mathcal{A}_n\right] = \widetilde{\mathbb{E}}\left[\frac{v_{n+1}(\mathbb{S}_{n+1})}{(1+r)^{n+1}} \,\middle|\, \mathbb{S}_n\right] . \tag{17}$$

This gives a difference equation for the functions $v_n(x)$ which can be determined iteratively by starting from the known function $v_N(x)$ and solving the above equation backwards.

Exercise 1a: (Option price as a function of stock price) Consider the four-period binomial model, i.e., $N = 4$. The function $v_4(x) = (x - K)^+$ is the payoff of the European call at the expiration time $N = 4$. Compute the functions v_3, v_2, v_1 and v_0 based on the above discussion. Take the coin toss probabilities for heads and tails as the risk-neutral probabilities $\tilde{p}$ and $\tilde{q}$ respectively. You can retain $\tilde{p}$, $\tilde{q}$ as they are without further substituting their expressions in terms of u, d and r.

Now consider the general N-period binomial model and guess the form of the functions $v_n(x)$ for $0 \le n \le N-1$ given $v_N(x)$. You can retain $v_N(x)$ as an unknown function, and in particular, you do not have to substitute the payoff $(x-K)^+$ of the European call at strike K. ■

We next go on to the numerical simulation of the multiperiod binomial model. We model a coin toss with probabilities p and q for heads and tails respectively using a random number generator as follows.

Exercise 1b: (Coin toss simulator) This exercise establishes a simple coin toss simulator based on the theoretical probability of generating a heads being p and of generating a tails is q, with $p + q = 1$.

A single coin toss is simulated by generating a single number x between $[0, 1]$ using a random number generator (e.g. `random.random` in Python). The outcome is a heads H if the the random number x lies between 0 and p ($0 \leq x \leq p$), and a tails T if x lies between p and 1 ($p < x \leq 1$). This is an example of the method of tower sampling used to sample discrete distributions.

Write code in your favorite programming language to implement this simple coin toss algorithm. It is recommended that you use the same programming language for the entire assignment.

Once you have the code to simulate a single coin toss, we can use it to simulate sequences of coin tosses. Write a program which takes as input the probabilities p, q, the number of coin tosses N, and the total number R of repeated trials of N coin tosses. We fix the value of N to be 120 for this exercise.

For a given p, q, the program should generate R sequences of $N = 120$ coin tosses using the coin toss generator that you have written, and it should also keep track of the number of heads that occur in each of the R repeated trials. You should get an array of R numbers corresponding to the number of heads in each of the R repeated trials. Create a histogram using these R values to obtain the distribution of the number of heads in the R trials. The output of the program should be this histogram. (You could also choose to output the array of R numbers and plot the histogram separately using another piece of code.)

Run the above program for three different choices of probabilities p and q:

1. $p = \frac{2}{3}$, $q = \frac{1}{3}$,
2. $p = q = \frac{1}{2}$,
3. $p = \frac{1}{4}$, $q = \frac{3}{4}$,

and for each choice of p, q, three different choices of the number R of repeated trials:

1. $R = 50$,
2. $R = 100$,
3. $R = 200$.

You should observe in each of the nine cases that the histogram is peaked around the value $pN = 120p$ corresponding to the fraction of the total number of coin tosses based on the theoretical probability of heads occurring being p. What happens to the shape of the histogram as R increases for a fixed p, q? ■

Exercise 1c: (Simulation of the stock price process) Next, we simulate the stock price process. Consider a particular trial of N coin toss sequences. Begin with the stock price $\mathbb{S}_0$. After the first coin toss, record the value of the stock $\mathbb{S}_1$ as $u\mathbb{S}_0$ if the outcome is a heads, and as $d\mathbb{S}_0$ if the outcome is a tails. Similarly, keep updating the stock value depending on the outcome of each subsequent coin toss till you reach N tosses. You will now have the final value of the stock $\mathbb{S}_N$ at the end of N coin tosses corresponding to a particular sequence of N generated randomly. Repeat the N coin toss simulation R times and obtain an array of R values of the final stock price $\mathbb{S}_N$ corresponding to the R repeated trials. Plot a histogram of the R final stock values.

This is an example of a Monte Carlo simulation of stock prices based on the binomial asset pricing model.

Take the initial stock price to be $\mathbb{S}_0 = 4$, and the up and down factors as $u = 2$, $d = \frac{1}{2}$. Take the number of coin tosses in each trial to be $N = 4$ for this exercise, that is, we are looking at the 4-period binomial model. You must obtain the histogram of final stock prices for the nine different cases considered in the previous exercise: three different choices of the coin toss probabilities and three different choices of the number R of repeated trials.

These histograms (after normalizing to unit area) estimate the true theoretical probability distribution for the final stock price values $\mathbb{P}\{\mathbb{S}_N = y\}$. The estimates become more reliable as the number R of repeated trials increases. ■

Exercise 1d: (Pricing a European call option) We know that the price of an option at time 0 is the expectation value

$$\mathbb{V}_0 = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N}\right] \;, \tag{18}$$

where $\widetilde{\mathbb{E}}$ is the expectation value taken with respect to the risk-neutral probabilities $\tilde{p}$, $\tilde{q}$ and $\mathbb{V}_N$ is the payoff of the option at time of expiry. For a European call option whose payoff at time N is a function only of the stock price $\mathbb{S}_N$ at time N, $\mathbb{V}_N = v_N(\mathbb{S}_N) = \max(\mathbb{S}_N - K, 0)$, the above formula can be written as

$$\mathbb{V}_0 = \widetilde{\mathbb{E}}\left[\frac{v_N(\mathbb{S}_N)}{(1+r)^N}\right] \;. \tag{19}$$

Let us take the initial stock price to be $\mathbb{S}_0 = 4$, up and down factors $u = d^{-1} = 2$, and work with a European call option with a slightly out-of-the-money strike price $K = 5$. Take the risk-free interest rate to be $r = \frac{1}{4}$ and the number of periods $N = 4$.

Recall from Assignment 2 that the expectation value above can be written either as a sum over elements in sample space or as a sum over the possible values of the final stock price:

$$\widetilde{\mathbb{E}}\left[\frac{v_N(\mathbb{S}_N)}{(1+r)^N}\right] = \sum_{\underline{\omega}\in\Omega_N} \frac{v_N(\mathbb{S}_N(\underline{\omega}))}{(1+r)^N}\widetilde{\mathbb{P}}\{\underline{\omega}\} = \sum_{y\in\mathrm{Ran}(\mathbb{S}_N)} \frac{v_N(y)}{(1+r)^N}\widetilde{\mathbb{P}}\{\mathbb{S}_N = y\} \;. \tag{20}$$

In the previous exercise, we concluded that the probability distribution $\mathbb{P}\{\mathbb{S}_N = y\}$ of final stock prices is reliably estimated by the normalized histogram of the final stock price values obtained for a large number of repeated trials of the N coin tosses. This implies that the above expectation value can also be reliably computed by simply taking the usual average of the numbers $(1+r)^{-N} v_N(\mathbb{S}_N)$ generated during the repeated trials. For concreteness, suppose the final stock price obtained in the i^{th} trial is designated y_i, with i running over 1 to R, where the coin tosses are generated using the risk-neutral probabilities. Then, the statement is

$$\mathbb{V}_0 = \widetilde{\mathbb{E}}\left[\frac{v_N(\mathbb{S}_N)}{(1+r)^N}\right] \approx \frac{1}{R}\sum_{i=1}^{R} \frac{v_N(y_i)}{(1+r)^N} \ . \tag{21}$$

That is, take the array of R final stock price values obtained during the R repeated trials, compute the array of the R final payoffs, and simply take the numerical mean of these R values to obtain a reliable estimate of the initial value $\mathbb{V}_0$ of the option.

Compute the initial price of the European call option with strike price $K = 5$ by running the Monte Carlo simulation for the following values of the number of repeated trials:

1. $R = 50$
2. $R = 100$
3. $R = 500$
4. $R = 1000$
5. $R = 10000$

What happens to the value of $\mathbb{V}_0$ that you obtain as you increase the number of repeated trials? ■

Exercise 1e: (Volatility dependence of option price) In this exercise, we explore the dependence of the initial option price on the up and down factors of the stock price process. For simplicity, we restrict ourselves to the symmetric case $u = d^{-1}$. Recall that higher the u, higher the volatility of the stock. Thus, we study the impact of the volatility of the stock on the fair option price.

Vary u from 2 to 3 in steps of 0.1 and run the Monte Carlo simulation of Exercise 1d to compute v_0 for each of these 11 values of u. Take the number R of repeated trials to be $R = 1000$. All other parameters are the same: the initial stock price is $\mathbb{S}_0 = 4$, we have a European call with strike price $K = 5$, the number of periods is $N = 4$ and the risk-free interest rate is $r = 25\%$.

Plot the $\mathbb{V}_0$ you obtain versus u. Do you see a trend here? If the trend is not clear from the 11 values you computed, take a finer step size, say 0.01 between 2 and 3 and run the simulation for these 101 values of u. You will have a much smoother curve in this case. ■

Exercise 1f: (Theoretical volatility dependence) Calculate the value $\mathbb{V}_0$ using the function $v_0(\mathbb{S}_0)$ obtained in Exercise 1a. Take $N = 4$, $r = 25\%$, initial stock price $\mathbb{S}_0 = 4$ and the European call's strike price $K = 5$. Check that the theoretical dependence on u

agrees with what you generated in Exercise 1e by plotting the option price as a function of u between 2 and 3. ■

Exercise 1g: (Time evolution of option price) In the four-period binomial model, calculate the value of the European call option at intermediate times using the conditional expectation formula

$$v_n(\mathbb{S}_n) = \widetilde{\mathbb{E}}\left[\frac{v_N(\mathbb{S}_N)}{(1+r)^{N-n}} \,\middle|\, \mathbb{S}_n\right], \quad \text{for } 1 \le n \le 3. \tag{22}$$

This is computed by the same procedure used in Exercise 1d but with some modifications to the program. Start with the initial stock price $\mathbb{S}_0 = 4$. First, record all possible values for the stock price $\mathbb{S}_n$ at time n by going through all possible paths in the binomial tree up to time n. For the four-period model, this can be done by hand or by a simple nested loop (in particular, there is no need for a Monte Carlo simulation for this step).

Now let us describe the main Monte Carlo simulation. Fix one of the allowed values of the stock price at time instant n and then run the Monte Carlo simulation with period $N-n$ and initial stock price equal to the fixed value $\mathbb{S}_n$, and then compute the mean of the discounted payoffs $(1+r)^{n-N}v_N(\mathbb{S}_N)$ over the R repeated trials. Repeat the above for every allowed value of $\mathbb{S}_n$. This will give the value v_n for every allowed value of $\mathbb{S}_n$.

The final set of values you must have are two values of v_1 corresponding to the two possible values of $\mathbb{S}_1$, three values of v_2 for the three allowed values of $\mathbb{S}_2$, and four values of v_3 for the four allowed values of $\mathbb{S}_3$. Check if these values agree with the values of the functions computed in Exercise 1a (for $u = d^{-1} = 2$, $K = 5$ and $r = 25\%$).

Take $R = 1000$ as in Exercise 1d. The other parameters are the same as in Exercise 1d: $u = d^{-1} = 2$, strike price of the European call $K = 5$, and risk-free interest rate 25%. ■

Exercise 1h: (Delta-hedge process) For the price evolution obtained above in Exercise 1g, calculate the Δ-hedge process

$$\mathbb{\Delta}_n(\mathbb{S}_n) = \frac{v_{n+1}(u\mathbb{S}_n) - v_{n+1}(d\mathbb{S}_n)}{u\mathbb{S}_n - d\mathbb{S}_n}, \tag{23}$$

where the values of v_{n+1} on the right hand side are the ones you found numerically in the previous exercise (and NOT the theoretical answers you got in Exercise 1a). ■

Exercise 1i: (Non-random drift) Let us go back to the stock price simulation in Exercise 1c, and consider non-symmetric evolution of stock prices, that is $u \ne d^{-1}$. We have two cases: $u > d^{-1}$ or $u < d^{-1}$. Let us choose $u = 1.2$, $d = 0.9$ for the first possibility and $u = 1.1$ and $d = 0.8$ for the second possibility. Choose the risk-free interest rate $r = 5\% = \frac{1}{20}$ to avoid arbitrage opportunities.

Consider $\mathbb{S}_0 = 4$, $N = 10$ and the number of repeated trials $R = 1000$. Run the Monte Carlo simulation for each choice of u, d above, and keep track of the stock price values at all time instants. Then, for each n, $1 \leq n \leq 10$, we have R values of the stock price $\mathbb{S}_n$ at time n. For each n, create a normalized histogram of the R values of $\mathbb{S}_n$. For a given choice of u, d, this procedure will give 10 histograms, one for each time instant n between 1 and 10. How do these histograms change as you change n? Do they move relative to each other or stay approximately in the same place? If they move, do they move to left or the right? (Optional: You can make a simple animated `gif` to show the time evolution of the 10 histograms for a quick visual way of conveying the drift.)

Now repeat the above simulation for $u = d^{-1} = \frac{4}{3}$, $r = 25\%$, $\mathbb{S}_0 = 4$, $N = 10$, $R = 1000$. How do the normalized histograms change as n increases from 1 to 10? ■

Exercise 1j: (Statistical drift) In this exercise, we work with $u = d^{-1} = \frac{4}{3}$, $r = 25\%$, $\mathbb{S}_0 = 4$, $N = 10$, $R = 1000$. However, we will consider different probabilities for the coin toss. Recall that the stock price process is a martingale if we use the risk-neutral probabilities $\tilde{p}$, $\tilde{q}$ which come out to be $1/2$ each for the above choice of u, d and r. The sequence of normalized histograms was plotted for this situation at the end of the previous exercise.

Now, obtain the 10 histograms for the stock prices for each time instant n between 1 and 10, but for different choices of probabilities:

1. $p = \frac{2}{3}$, $q = \frac{1}{3}$
2. $p = \frac{1}{4}$, $q = \frac{3}{4}$

Again, comment on the movement of the normalized histograms in each of the above cases, and compare the movement to the risk-neutral case. Which of these is a submartingale and which of these is a supermartingale? Apart from moving to the left or right, what are the changes in the shape of the histograms? How are these changes different compared to the non-random drift of Exercise 1i? ■

Not for assessment: Simulate the *Asian* call option given in Exercise 1.8 of Chapter 1, Shreve, Part I, and the *lookback* option in Example 1.2.4 in Chapter 1, Shreve, Part I. These are path-dependent options, which means we have to use the original formula (4)

$$\mathbb{V}_0 = \widetilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N}\right] , \tag{24}$$

directly, without using any specific Markov property. This involves only a minor modification of your programs in the previous exercises where we simulated a European call option. In particular, one has to again take the simple numerical mean of the discounted final payoffs that one obtains in the R repeated trials. Since this exercise is not evaluation, you can play around with different values of $\mathbb{S}_0$, u, d, r and N. It is recommended that you run at least $R = 1000$ repeated trials.